

Introduction

Schizophrenia and related psychotic disorders are characterised by the presence of positive, negative and disorganised symptoms, each having severe impacts on patients' lives. Although negative and disorganised symptoms have been well associated with genetic and clinical predictors, little is known about what predicts severity of positive symptoms, and therefore, psychosis. Disentangling the heterogeneity of positive symptoms could give insight into symptom-specific predictors, and the mechanisms underlying these conditions.

Objective

To investigate the association between clinical, social and genetic predictors of psychosis and the severity of individual psychotic symptoms in a cross-sectional clinical cohort.

Methods

1297 participants from CardiffCOGs, a cross-sectional clinical sample were analysed. Lifetime symptom severity was measured using the Scale for Assessment of Positive Symptoms (SAPS). Clinical and social predictors were assessed through clinical interviews and validated questionnaires. Polygenic risk scores for schizophrenia, bipolar disorder, ADHD, autism and intelligence were calculated from blood samples. Univariate linear regressions with age and sex as covariates were used to test the association between each of the different behavioural, cognitive and genetic predictors (15 in total) and the severity of individual positive symptoms (20 in total). Multiple testing was controlled using a False Discovery Rate analysis.

Results

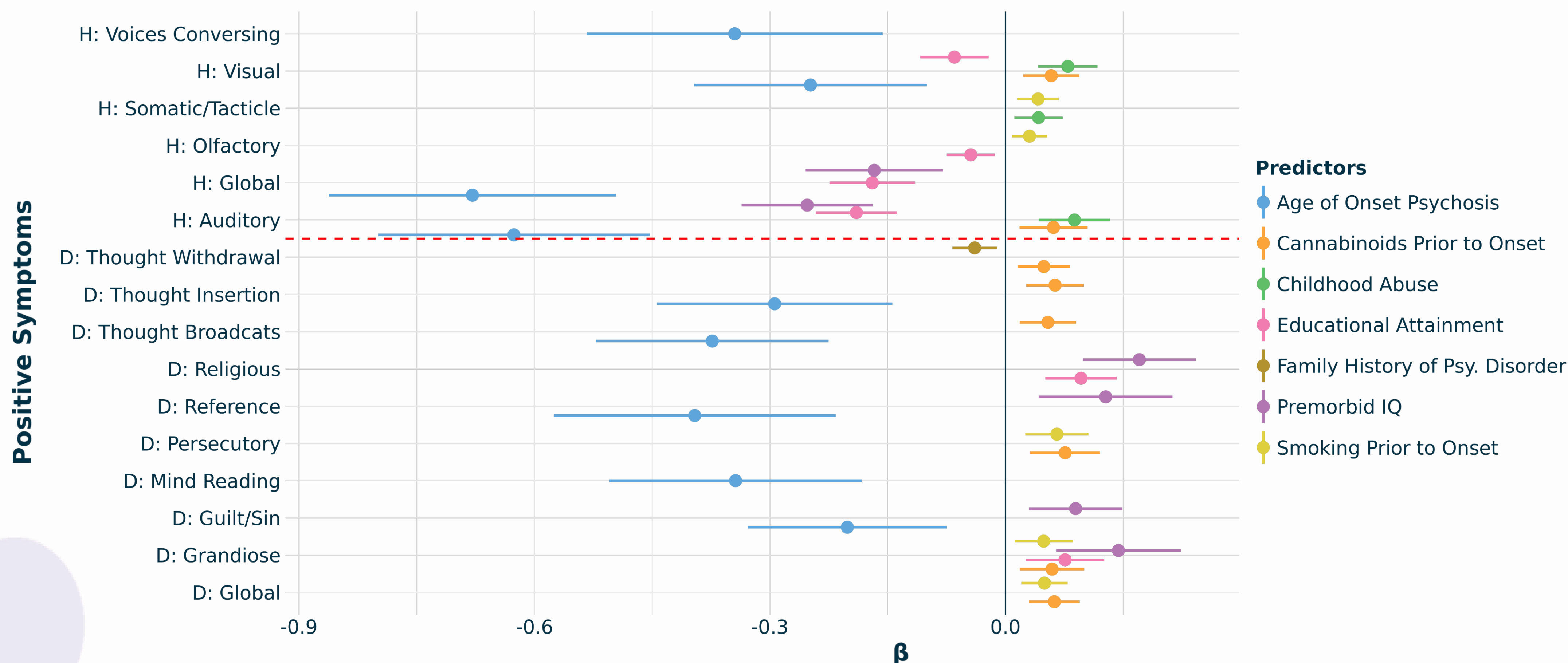


Figure 1: Significant associations between different predictors and individual psychotic symptoms. Points indicate normalised β estimates, and horizontal lines represent 95% confidence intervals. Hallucinations were labelled with an "H" and were placed below the red line. Delusions are labelled with a "D" and are below the red line.



Earlier age of onset, cannabis use and tobacco smoking before onset of psychosis were found to be general predictors of severity of many positive symptoms.



Genetic liability to schizophrenia or other mental health conditions **DID NOT** strongly predict the severity of positive symptoms.



Premorbid IQ was specifically associated with higher severity of religious, grandiose, referential and guilt-based delusions, but negatively associated with global and auditory hallucinations. There was a similar pattern of findings for educational attainment.



Previous experience of childhood abuse showed a specific effect on severity of hallucinations.

Conclusions

The present findings suggest that while certain predictors indicate general severity of positive symptoms in psychosis, others have symptom-specific effects. Consistent with the recent models of psychosis, these results lend empirical insight into the factors underlying hallucinations and delusions formation, as well as the heterogeneity of symptoms within psychotic disorders.

¹Cardiff University, School of Psychology, Cardiff, United Kingdom

²Cardiff University, Centre for Neuropsychiatric Genetics and Genomics, Division of Psychological Medicine and Clinical Neurosciences, School of Medicine, Cardiff, United Kingdom